KATO SAMUEL

Full-Stack Software Engineer | Flutter Mobile Developer | Backend Engineer

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Engineer and Mobile Developer completed a Bachelor of Business Computing (Hons) at Makerere University Business School (MUBS), Uganda , expected graduation June 2026. Experienced in designing, containerising and deploying scalable web and mobile applications using NestJS, Node.js, Next.js, React, Flutter and Laravel, with strong database expertise in PostgreSQL, Prisma ORM, MySQL and Supabase. Delivered production-grade systems across education, property management, finance, agriculture and retail, including live VPS deployments with Docker, Traefik/Nginx, GitHub Actions CI/CD and Let's Encrypt. Skilled in REST API design, multi-tenant SaaS architecture, server-side role-based access control, background job processing with Redis/BullMQ, and Linux server administration.

SKILLS

Frontend:	Next.js 15/16, React 19, TypeScript, JavaScript, Tailwind CSS 4
Backend:	NestJS, Node.js, Laravel, PHP, REST APIs, microservices-ready design
Mobile:	Flutter, Dart, Offline-first Architecture, Hive, Supabase (auth + real-time sync)
Databases:	PostgreSQL, MySQL, Prisma ORM, Supabase, RLS (row-level security)
DevOps:	Docker, Docker Compose, GitHub Actions CI/CD, Traefik, Nginx, Linux (Ubuntu), Vercel
Auth / Security:	JWT, Argon2, Laravel Sanctum, RBAC, Spatie Permissions, Helmet, Rate Limiting
Jobs & Media:	Redis / BullMQ queues, AWS S3 presigned URLs, Cloudinary, Nodemailer, Resend
Practices:	Agile, Scrum, MVC, Git / GitHub, clean architecture, documentation

Keywords: Full Stack Development, Software Engineering, Mobile Development, REST APIs, Microservices, Scalable Systems, Cloud Deployment, Agile Development, NestJS, Next.js Developer, Laravel Developer, Flutter Developer, PostgreSQL, MySQL, Multi-Tenant SaaS, CI/CD

PROJECT EXPERIENCE

SchoolOS — Multi-Tenant School Management Platform

NestJS 10, Node.js, Next.js 15, React 19, TypeScript, Prisma ORM, PostgreSQL (149 models, RLS), Redis/BullMQ, AWS S3, Docker

Architected a pnpm/Turborepo monorepo combining a 50+ module NestJS REST API (Swagger-documented) with a Next.js 15 App Router / React 19 frontend — a multi-tenant platform served across three strictly separated surfaces: public website, private per-school portal, and internal platform admin console.

Designed server-side RBAC with 100+ granular permission keys resolved from Role/Permission tables and enforced in middleware, layouts and routes, so one school can never see another school's data and customers can never reach platform control features.

Built on Prisma 5 + PostgreSQL (149 models, RLS-enforced), JWT auth with Argon2 hashing and rotating refresh cookies, BullMQ/Redis async workers with a dedicated worker process, S3 presigned-upload media pipeline, and content processing (PDF/DOCX/XLSX/CSV/images).

- Delivered a CBC component-based assessment, marks and results engine plus a template-driven report-card system producing customisable A4 report cards with immutable published snapshots and version history.

Live: https://schoolos.conquerer.org

Tica — Property Management ERP

NestJS 11, TypeScript, Prisma, PostgreSQL 15, Next.js 14, Docker Compose, Nginx + Let's Encrypt, GitHub Actions

- Designed a modular NestJS REST API with 19 domain modules (owners, properties, units, tenants, occupancies, billing, payments, arrears, commission, expenses, issues, reminders, reports, audit, settings) backed by Prisma/PostgreSQL.
- Engineered a cron-driven monthly billing engine with arrears aging, a payment-allocation engine (pre-takeover vs company-managed obligations) and a commission engine where owner statements derive only from admin-verified transactions — with an append-only audit log and per-owner data isolation.
- Deployed as a Docker Compose stack behind an Nginx reverse proxy with Let's Encrypt SSL and a GitHub Actions deploy pipeline; added JWT auth, rate limiting and fine-grained role/permission guards.

GitHub: github.com/katosamuelraymond/Tica-Property-management *(private repo)*

samueltechug — Self-Hosted Portfolio (Live)

Next.js 16, React 19, TypeScript, Tailwind CSS 4, Framer Motion, Resend, Docker, Traefik, Let's Encrypt, GitHub Actions

- Deployed a production Next.js/React site to a self-hosted Ubuntu VPS using Docker with Traefik reverse proxy, automatic Let's Encrypt TLS and www→apex 301 redirects — the whole stack own and operated end-to-end.
- Built GitHub Actions CI/CD that publishes the container to GHCR and deploys to production and staging environments with Docker health-check and automatic rollback on failure.
- Animated marketing UI with Framer Motion and email contact forms wired through Resend server actions.

Live: https://samueltechug.dev

GitHub: github.com/katosamuelraymond/samueltechug

Creative Cube — Furniture E-Commerce Platform (Live)

Next.js 16, React 19, TypeScript, Prisma 7, PostgreSQL, Cloudinary, Tailwind CSS 4, Docker

- Built a production full-stack e-commerce platform with a sleek storefront and a comprehensive admin dashboard for product, category, cart and order management, organised as a clean service layer (product/category/cart/order services).
- Implemented secure JWT authentication (jose), zod-validated forms, bcryptjs password hashing, a normalised Prisma/PostgreSQL schema with migrations and seed data, and Cloudinary media/CDN handling.

- Automated containerised deployments to a VPS via GitHub Actions with health-checked rollback and a Prisma migrate step for staging.

Live: https://next.conquerer.org

GitHub: github.com/katosamuelraymond/creative-cube-next

Fintrack — Personal Finance Mobile App

Flutter, Dart, Supabase (PostgreSQL), Provider, SharedPreferences

- Built a cross-platform finance tracker with income/expense entry, budget tracking, notifications and chart-based financial reports.
- Designed a Supabase schema with per-user row isolation (RLS-aware transactions/notifications tables linked to auth.users) and real-time data synchronisation backed by Supabase Auth.

GitHub: github.com/katosamuelraymond/Fintrack

AgroLink — Offline-First Agriculture Supply Chain

Flutter, Dart, Hive (NoSQL), Offline-first Architecture

- Engineered an offline-first Flutter app (no internet required) connecting farmers, buyers and transporters in low-connectivity rural Uganda — aimed at eliminating middlemen and reducing post-harvest losses.
- Delivered three role-specific workflows — produce listing and order management, transparent buyer pricing with MTN MoMo / Airtel Money / cash payments, and pickup → delivery logistics tracking — with typed Hive persistence in isolated data boxes.

GitHub: github.com/katosamuelraymond/agrolink

POS System — Enterprise POS & ERP (Live)

Laravel, React (Vite), MySQL, Laravel Sanctum, Spatie Permissions, TanStack Query, Zustand

- Built a production POS/ERP covering sales, real-time inventory, customers and financial reporting with multi-branch support and offline capability.
- Implemented secure token authentication with Laravel Sanctum and fine-grained RBAC with Spatie permissions across a decoupled React SPA using TanStack Query and Zustand for state management.

Live: https://pos-staging.conquerer.org/home

EDUCATION

Bachelor of Business Computing (Hons) *— 2023-2026*

Makerere University Business School (MUBS), Kampala, Uganda

Focus: Business Systems Analysis, Software Development, Database Systems, Enterprise Computing

PROFESSIONAL ATTRIBUTES

- Strong analytical and problem-solving skills with the ability to translate business requirements into reliable, scalable software solutions.
- Proven experience delivering end-to-end production systems across education, property management, finance, agriculture and retail — including live, self-hosted deployments.

- Skilled in NestJS modular architecture, multi-tenant SaaS designs, background jobs with Redis/BullMQ, and scalable PostgreSQL/Prisma data models.
- Committed to code quality, security (authN/authZ, RLS, input validation), performance optimisation and documentation best practices.
- Self-motivated team player who rapidly learns and applies emerging technologies in fast-paced environments.

DECLARATION

I declare that the information provided in this curriculum vitae is true and accurate to the best of my knowledge. I am available for an interview at your earliest convenience.

REFEREES

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